Applications: Growth and Decay

ANSWER KEY

Section 1: Population Growth Models

1 A bacterial culture doubles every 3 hours. If the initial count is 500, what is the population after 9 hours?

a. 4000 ✓

b. 1500

c. 2000

d. 8000

After 9 hours, $9/3 = 3$ doubling periods have elapsed. $P(9) = 500 \times 2^3 = 500 \times 8 = 4000$.

2 A city has population $P(t) = 100000e^{0.02t}$, where t is in years. Find the continuous annual growth rate and doubling time.

a. **Rate** $= 2\%$**, Doubling time** ≈ 34.7 years ✓

b. Rate $= 20\%$, Doubling time ≈ 3.5 years

c. Rate $= 0.02\%$, Doubling time ≈ 50 years

d. Rate $= 2\%$, Doubling time ≈ 50 years

Growth constant $k = 0.02 = 2\%$. Doubling time $T = \frac{\ln 2}{k} = \frac{0.69315}{0.02} \approx 34.66 \approx 34.7$ years.

3 If a culture grows from 200 to 600 in 4 hours according to $P(t) = P_0 e^{kt}$, find k.

a. $k = \frac{\ln 3}{4} \approx 0.275$ ✓

b. $k = \frac{\ln 4}{3}$

c. $k = 3/4$

d. $k = \ln(3/4)$

$600 = 200e^{4k} \Rightarrow e^{4k} = 3 \Rightarrow 4k = \ln 3 \Rightarrow k = \frac{\ln 3}{4} \approx 0.275$.

Section 2: Radioactive Decay and Half-Life

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4 Carbon-14 has a half-life of 5730 years. What fraction of the original C-14 remains in a fossil after 17190 years?

a. $1/8$ ✓

b. $1/4$

c. $1/16$

d. $1/3$

Number of half-lives $= 17190/5730 = 3$. Remaining fraction $= (1/2)^3 = 1/8$.

5 State the formula relating decay constant $k > 0$ and half-life $t_{1/2}$ in $N(t) = N_0 e^{-kt}$.

a. $k = \frac{\ln 2}{t_{1/2}}$ ✓

b. $k = \frac{t_{1/2}}{\ln 2}$

c. $k = \frac{1}{2t_{1/2}}$

d. $k = \ln(2t_{1/2})$

$N_0/2 = N_0 e^{-kt_{1/2}} \Rightarrow e^{kt_{1/2}} = 2 \Rightarrow k = \frac{\ln 2}{t_{1/2}}$.

6 A sample contains 80 mg of an isotope with a half-life of 10 days. How much remains after 25 days?

a. $80(0.5)^{2.5} \approx 14.14$ mg ✓

b. 20 mg

c. 10 mg

d. 15 mg

$N(25) = 80(1/2)^{25/10} = 80(1/2)^{2.5} = \frac{80}{4\sqrt{2}} = \frac{20}{\sqrt{2}} = 10\sqrt{2} \approx 14.14$ mg.

Section 3: Newton's Law of Cooling

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7 State Newton's Law of Cooling formula for temperature $T(t)$ with ambient medium T_m.

a. $T(t) = T_m + (T_0 - T_m)e^{-kt}$ ✓

b. $T(t) = T_0 e^{-kt} + T_m$

c. $T(t) = T_m(1 - e^{-kt})$

d. $T(t) = (T_0 - T_m)e^{kt}$

The rate of cooling is proportional to the temperature difference:
$\frac{dT}{dt} = -k(T - T_m) \Rightarrow T(t) = T_m + (T_0 - T_m)e^{-kt}$.

8 A hot cup of coffee at $190\,^\circ\text{F}$ is placed in a $70\,^\circ\text{F}$ room. What is the initial temperature difference $T_0 - T_m$?

a. $120\,^\circ\text{F}$ ✓

b. $190\,^\circ\text{F}$

c. $70\,^\circ\text{F}$

d. $260\,^\circ\text{F}$

$T_0 - T_m = 190 - 70 = 120\,^\circ\text{F}$.

9 If the coffee cools from $190\,^\circ\text{F}$ to $130\,^\circ\text{F}$ in 10 minutes in the $70\,^\circ\text{F}$ room, find k.

a. $k = \frac{\ln 2}{10} \approx 0.0693 \text{ min}^{-1}$ ✓

b. $k = 0.1$

c. $k = \frac{\ln 3}{10}$

d. $k = 0.05$

$130 = 70 + 120e^{-10k} \Rightarrow 60 = 120e^{-10k} \Rightarrow e^{-10k} = 1/2 \Rightarrow 10k = \ln 2 \Rightarrow k = \frac{\ln 2}{10}$.

Section 4: Logarithmic Scales (pH, Richter, Decibels)

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10 The pH of a solution is defined as $\text{pH} = -\log_{10}[\text{H}^+]$. What is the pH if $[\text{H}^+] = 10^{-4}$ M?

a. 4 ✓

b. -4

c. 10

d. 14

$\text{pH} = -\log_{10}(10^{-4}) = -(-4) = 4$.

11 The Richter magnitude scale is $M = \log_{10}\left(\frac{I}{S}\right)$. How many times more intense is an earthquake of magnitude 7 than one of magnitude 4?

a. 1000 **times (**10^3**)** ✓

b. 3 times

c. 30 times

d. 10000 times

Difference in magnitude $\Delta M = 7 - 4 = 3$. Ratio of intensities $= 10^3 = 1000$.

12 Sound intensity level in decibels is $\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$. If the intensity I is multiplied by 100, how many decibels are added?

a. $+20$ dB ✓

b. $+100$ dB

c. $+10$ dB

d. $+2$ dB

$\Delta\beta = 10\log_{10}(100) = 10(2) = 20$ dB.

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